GREG FURLICH

Research Associate (Tenure-track Equivalent)

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Research Faculty and the Space Domain Awareness research lead for the Center for National Security Initiatives at the University of Colorado Boulder. Leads the Data Exploitation Lab for Trusted Autonomy (DELTA) research group. Research interests include sensors and sensing, mission data processing, and trusted autonomy. PhD in Physics from the University of Utah with a thesis on the ultraviolet remote sensing of ultra-high energy cosmic ray interactions within the atmosphere. Raised over $2.63 M in awards in the past 4 years and conferred a affiliation with the Aerospace Engineering Sciences at CU Boulder.

EDUCATION

Doctorate of Philosophy, *Physics* 2014 – 2020
Master of Science, *Physics* 2014 – 2018
University of Utah, Salt Lake City, UT

Bachelor of Science, *Physics* 2010 – 2014
Michigan Technological University, Houghton, MI *Magna Cum Laude*
Minors: Mathematical Sciences and German

RESEARCH INTERESTS

Sensors and Sensing: image collection, processing, and analysis over many spectral regimes (visible, infrared, ultraviolet, multispectral, hyperspectral); expeditionary low SWAP sensor systems for situational awareness; target signature phenomenologies

Mission Data Processing: signal, image, and video processing; feature extraction and exploitation; algorithm development for dim object detection and disparate data fusion for object tracking and state estimation; event detection, characterization, and typing; machine learning for automatic target recognition, stochastic processing modeling, anomaly detection, physics-informed neural networks

Trusted Autonomy: autonomous decisions for proliferated, distributed, and decentralized systems; battle management architectures; killchain or threat decomposition; course of action recommendation; hostility and intent assessment; risk assessment and management

PROFESSIONAL RESEARCH EXPERIENCE

Research Associate Aug 2022 – Present
Space Domain Awareness Research Lead
University of Colorado Boulder,
Center for National Security Initiatives (NSI),
Data Exploitation Lab for Trusted Autonomy (DELTA)

- Leading the Data Exploitation Lab for Trusted Autonomy (DELTA) research group, a multidisciplinary group focused on applied physics, mathematics, and computer science for mission data processing for SDA, MWD, and ISR.
- Led many multi-disciplanary DoW and IC proposal efforts as CU Principal Investigator. Awarded $2.63 M in funding efforts on grant proposals and contracts.
- Led applied research projects in the USSF Space Systems Command (SSC) Boulder and Space Domain Awareness (SDA) Tools, Applications, and Processing (TAP) Labs. Selected for the inaugural Project Apollo Technology Accelerator cohort of the SDA TAP Lab and continued participation up to Cohort 9 before the SDA TAP Lab became the BMC3I TAP Lab.
- Developed and successfully tested a unrestricted operational capability for detection of launches using publicly available weather satellite imagery. Capability is named Open-data Weather imagery Launch Alerts and Tracklets (OWLAT).
- Selected as a fellow for the AFOSR Summer Faculty Fellowship Program and invited to perform impactful research at the AFRL Information Directorate (AFRL/RI).

PROFESSIONAL RESEARCH EXPERIENCE CONTINUED

A/AI Research Engineer — May 2022 – Aug 2022
Senior Research Scientist — Jan 2021 – May 2022

Lockheed Martin Space Systems,
Advanced Programs and Exploitations (APEX)

- Developed innovative algorithms which improved the efficiency and accuracy in signal and image processing for a variety of sensor types (visible, infrared, multispectral, and RF). Developed data fusion algorithm for object detection, tracking, and state estimation.
- Designed novel machine learning networks for super resolution, image-to-image translation, and style transfer with Generative Adversarial Networks (GANs), image segmentation with UNETs and FCNs, and object classification with Convolution Neural Networks (CNNs).
- Selected as a Lockheed Martin Recognized Technical Talent for technical contributions within first year.
- Generated intellectual property which provided technical advantage for many applications and was therefore awarded and protected as a trade secret.

Graduate Research Assistant in Cosmic Rays — 2014 – 2020

Telescope Array (TA) Cosmic Ray Observatory, Institute of High Energy Astrophysics,
Department of Physics and Astronomy, University of Utah

Research Advisor: Douglas Bergman

Thesis : Observation of the GZK Suppression with the Telescope Array Fluorescence Telescopes and Deployment of the Telescope Array Expansion

- Analyzed 10 years of monocular fluorescence data to create a Cosmic Ray Energy Spectrum. Raw cosmic ray event data were analyzed by reconstructing event geometry and energy. The event information was aggregated into a large database to produce the event energy distribution. The aperture of the detectors in monocular mode was calculated through Monte Carlo thrown events over TA and simulating the response of the detectors.
- Analyzed the significance of the Greisen–Zatsepin–Kuzmin (GZK) Suppression at the highest energies of cosmic rays in the 10 year cosmic ray spectrum. Analyzed other features that give insight into cosmic ray populations or propagation effects.
- Classified weather over TA with the fluorescence detector photomultiplier tube baselines to create false color images of the detectors' field of view. Videos of these snapshots were then used as inputs in a Recurrent Convolution Neural Network (RCNN) model constructed with Keras machine learning framework to determine clear weather data for further cosmic ray analysis.
- Refurbished, tested, calibrated, and deployed the system of ultraviolet fluorescence telescopes for the TAx4 expansion. Collected first light with these new instruments.
- Analyzed the preliminary data from the new fluorescence detectors for the expansion of TA known as TAx4 into a preliminary Cosmic Ray Energy Spectrum and compared to previous results to gauge the new detector's capabilities.

AWARDS

Advanced Maui Optical and Space Surveillance Technology Conference (AMOS) Best Poster	Sept 2025
Air Force Office of Scientific Research (AFOSR) Summer Faculty Fellow, AFRL/RI	Summer 2023
Recognized Technical Talent, Lockheed Martin	Selected 2021
Departmental Scholar, Department of Physics, Michigan Technological University	2013
Sigma Pi Sigma, Physics Honor Society	Inducted 2013
Michigan Space Grant Consortium Recipient	2012

PUBLICATIONS

Reinforcement Learning Agent-Based Multi-Target Tracking with Closely-Spaced Objects (CSO), S. Rossland, B. Sweeney, T. Gill, **G. Furlich**, B. Oh, J. Fraser, M. W. Chan, S. Kerner, IEEE Aerospace Conference, 2027, *Accepted*

Time-resolved leader spectra of downward Terrestrial Gamma-ray Flashes observed at the Telescope Array Surface Detector, N. Kieu et al., *Journal of Geophysical Research: Atmospheres*, 130, e2025JD043812, 2025, https://doi.org/10.1029/2025JD043812

Intermediate fluence downward terrestrial gamma-ray flashes as observed by the Telescope Array Surface Detector, Abbasi, R. et al., *Journal of Geophysical Research: Atmospheres*, 129, e2024JD041260, 2025, https://doi.org/10.1029/2024JD041260

PUBLICATIONS CONTINUED

First Time-Resolved Leader Spectra Associated With a Downward Terrestrial Gamma-Ray Flash Detected at the Telescope Array Surface Detector, N. Kieu et al., *Journal of Geophysical Research: Atmospheres*, 129, e2024JD041720, 2024, https://doi.org/10.1029/2024JD041720

Isotropy of cosmic rays beyond 10^{20} eV favors their heavy mass composition, R.U. Abbasi et al. (The Telescope Array Collaboration), Phys. Rev. Lett., **133**, 041001, 2024, doi: 10.1103/PhysRevLett.133.041001

Mass composition of ultrahigh energy cosmic rays from distribution of their arrival directions with the Telescope Array, R.U. Abbasi et al. (The Telescope Array Collaboration), Phys. Rev. Lett. **110**, 022006, 2024, doi: 10.1103/PhysRevD.110.022006

An extremely energetic cosmic ray observed by a surface detector array, R.U. Abbasi et al. (The Telescope Array Collaboration), Science, **382**, 2023, doi: 10.1126/science.abo5095

The energy spectrum of cosmic rays measured by the Telescope Array using 10 years of fluorescence detector data, R.U. Abbasi et al. (The Telescope Array Collaboration), Astropart. Phys., **151**, 2023, doi: 10.1016/j.astropartphys.2023.102864.

Infrared Small Target Detection Enhancement Using a Lightweight Convolutional Neural Network, M. Gupta, J. Chan, M. Krouss, **G. Furlich**, P. Martens, M. Chan, M. L. Comer, E. J. Delp, Geoscience and Remote Sensing Letters, 2022, doi: 10.1109/LGRS.2022.3203931.

Search for Spatial Correlations of Neutrinos with Ultra-high-energy Cosmic Rays, A. Albert et al. (The ANTARES Collaboration, the IceCube Collaboration, the Pierre Auger Collaboration, and the Telescope Array Collaboration), 2022 ApJ **934** 164. DOI 10.3847/1538-4357/ac6def

Observation of variations in cosmic ray single count rates during thunderstorms and implications for large-scale electric field changes, R. U. Abbasi et al. (The Telescope Array Collaboration), Phys. Rev. D **105** 164, 2022. DOI: 10.1103/PhysRevD.105/062002

The Cosmic-Ray Composition between 2 PeV and 2 EeV Observed with the TALE Detector in Monocular Mode, R. U. Abbasi et al. (The Telescope Array Collaboration), 2021 ApJ **909** 178. DOI 10.3847/1538-4357/abdd30

Measurement of the proton-air cross section with Telescope Array's Black Rock Mesa and Long Ridge fluorescence detectors, and surface array in hybrid mode, R. U. Abbasi et al. (The Telescope Array Collaboration), Phys. Rev. D **102**, 062004, 2020. DOI: https://doi.org/10.1103/PhysRevD.102.062004

Evidence for a Supergalactic Structure of Magnetic Deflection Multiplets of Ultra-high-energy Cosmic Rays, R. U. Abbasi et al. (The Telescope Array Collaboration), 2020 ApJ **899** 86, 2020. DOI 10.3847/1538-4357/aba26c

Search for Ultra-High-Energy Neutrinos with the Telescope Array Surface Detector, R. U. Abbasi et al. (The Telescope Array Collaboration), J. Exp. Theor. Phys. 131, 255–264 (2020). https://doi.org/10.1134/S106377612005012X

Observations of the Origin of Downward Terrestrial Gamma-Ray Flashes, JW Belz et al., Journal of Geophysical Research: Atmospheres, 125, e2019JD031940, 2020. https://doi.org/10.1029/2019JD031940

Observation of the GZK Suppression with the Telescope Array Fluorescence Telescopes and Deployment of the Telescope Array Expansion, **Greg Furlich**, Thesis, University of Utah, April 2020

Constraints on the diffuse photon flux with energies above $10^{18}eV$ using the surface detector of the Telescope Array experiment, R.U. Abbasi et al. (The Telescope Array Collaboration), Astropart. Phys. **110** (2019)

Testing a Reported Correlation between Arrival Directions of Ultra-high-energy Cosmic Rays and a Flux Pattern from nearby Starburst Galaxies using Telescope Array Data, R.U. Abbasi et al. (The Telescope Array Collaboration), ApJ **867** 2 (2018)

The Cosmic-Ray Energy Spectrum between 2 PeV and 2 EeV Observed with the TALE detector in monocular mode, R.U. Abbasi et al. (The Telescope Array Collaboration), ApJ **865** 74 (2018)

Evidence of Intermediate-scale Energy Spectrum Anisotropy of Cosmic Rays $E \geq 10^{19.2}eV$ with the Telescope Array Surface Detector R.U. Abbasi et al. (The Telescope Array Collaboration), ApJ **862** 91 (2018)

Depth of Ultra High Energy Cosmic Ray Induced Air Shower Maxima Measured by the Telescope Array Black Rock and Long Ridge FADC Fluorescence Detectors and Surface Array in Hybrid Mode, R.U. Abbasi et al. (The Telescope Array Collaboration), ApJ **858** 76 (2018)

Gamma-ray Showers Observed at Ground Level in Coincidence With Downward Lightning Leaders, R.U. Abbasi et al. (The Telescope Array Collaboration), Journal of Geophysical Research: Atmospheres **123** (2018)

PROCEEDINGS AND NON-REVIEWED PUBLICATIONS

Transformer-Based Resident Space Object (RSO) Characterization from Photometric Time Series Observations, S Rossland, **G. Furlich**, Tanner Gill, Brian Sweeney, 2026 Advanced Maui Optical and Space Surveillance Technology Conference (AMOS), *Proceeding Forthcoming*

Tanner Gill, **Greg Furlich**, Nisar Ahmed. 2026. "Multi-Phase Guidance and Control for Small Satellite Docking via Deep Reinforcement Learning," Proceedings of the Small Satellite Conference, Poster Session, SSC26-P1-48, https://digitalcommons.usu.edu/smallsat/2026/all2026/140/, DOI: https://doi.org/10.26077/zcd1-fy48

Angela Crews et al. 2026. "Analysis of Satellite Drag, Orbit Tracking, and Associated Operational Impacts in Very Low Earth Orbit (VLEO)," Proceedings of the Small Satellite Conference, Poster Session, SSC26-RAV-01, https://digitalcommons.usu.edu/smallsat/2026/all2026/25/, DOI: https://doi.org/10.26077/g98e-gg07

A Spatial-Spectral Convolutional Neural Network for Dim, Unresolved Target Detection in Multispectral Imagery, B Sweeney, **G Furlich**, S Rossland, M W Chan, S Kerner, SPIE Defense + Security, 2026

Data-driven chromaticity-space rotations for enhanced target detection in multispectral overhead imagery, S Rossland, **G Furlich**, M W Chan, S Kerner, SPIE Defense + Security, 2026

In-Transit Space Object Recognition via Physics-Based Remote Sensing, B Joens, K Miller, **G Furlich**, A Shirkhodaie, J Christopher, V Sheffey, SPIE Defense + Security, 2026

Multi-modal physics simulation for space launch vehicle awareness and monitoring, B Joens, K Miller, **G Furlich**, A Shirkhodaie, J Christopher, V Sheffey, SPIE Defense + Security, 2026

Utilizing Civilian Launches as Live Exercises for Evaluating a Federated Protect and Defend SDA Battle Management System, **G. Furlich** et al., 2025 Advanced Maui Optical and Space Surveillance Technology Conference (AMOS) (**Best Poster Award**)

Towards an AI-enabled Space Battle Management System on Space Protocol's Quantum Resilient Blockchain, Latif, Yasir ; Bagchi, Samya ; Madhuri Pratti, Latha ; Reed, Harvey ; Stilwell, Ruth ; **Furlich, Greg**, 2025 Advanced Maui Optical and Space Surveillance Technology Conference (AMOS)

Automated, Collaborative Applications to Close Kill Chain Gaps, **G. Furlich** et al., 2024 Advanced Maui Optical and Space Surveillance Technology Conference (AMOS)

SubPixel Localization of Objects Using Multiple Spectral Bands, M. Gupta, J. Chan, M. Krouss, **G. Furlich**, P. Martens, M. Chan, M. L. Comer, E. J. Delp, IEEE Aerospace Conference, 2022

Recent measurement of the Telescope Array energy spectrum and observation of the shoulder feature in the Northern Hemisphere, D. Ivanov, D. Bergman, **G. Furlich**, R. Gonzalez, G. Thomson and Y. Tsunesada, Proceedings of Science 395 (ICRC2021), 341, 37th International Cosmic Ray Conference, Berlin, Germany, July 2021

Telescope Array 10-Year Monocular Spectrum, Douglas Bergman, **Greg Furlich**, Proceedings of Science 395 (ICRC2021), 339, 37th International Cosmic Ray Conference, Berlin, Germany, July 2021

Telescope Array FD Weather Classification using Machine Learning, **Greg Furlich**, Proceedings of Science (ICRC2019) 261, 36th International Cosmic Ray Conference, Madison, WI, July 2019

Towards a Telescope Array 10 Year FD Monocular Energy Spectrum, **Greg Furlich**, Douglas Bergman, Proceedings of Science (ICRC 2019) 260, 36th International Cosmic Ray Conference, Madison, WI, July 2019

Increasing the Yield of Boron Nitride Nanotubes in a CVD Method, **Greg Furlich**, Michigan Space Grant Consortium, Michigan in Space **14** (2012)

PRESENTATIONS

Reinforcement Learning Agent Based Multi-Target Tracking with Closely Spaced Objects (CSOs), IEEE Aeroconf 2027, Big Sky, MT, *Accepted*

Applied Astronomy: Using Astronomy Methods for Ground or Space-based Remote Sensing of Anthropogenic Objects in Space, Michigan Technological University (MTU) Physics Semimar, Houghton, MI, April 2026

Data Exploitation for Space Domain Awareness Mission Data Processing, AES Department Seminar, University of Colorado Boulder, Boulder, CO, January 2026

Open-data Weather imagery Launch Alerts and Tracklets, AFRL Blank Space Talk, Virtual, January 2026

AMOS Encore, Expert Talk, SDA TAP Lab, January 2026

Defeating Emerging Threats with Advanced Algorithms, United States Space Command (USSPACECOM) Space Concepts 2040, Colorado Springs, CO, April 2024

PRESENTATIONS CONTINUED

Defeating Emerging Threats with Advanced Algorithms, United States Space Command (USSPACECOM) Academic Engagement Enterprise (AEE), Colorado Springs, CO, April 2024

Multi-spectral Object Detection Algorithms, Joint Algorithm Workshop, Lockheed Martin, Orlando, FL, April 2022

Observation of the GZK Suppression with the Telescope Array Fluorescence Telescopes and Deployment of the Telescope Array Expansion, Thesis Defense, University of Utah, April 2020

Weather Classification using Machine Learning at the Telescope Array Cosmic Ray Observatory, APS April Meeting, Denver, CO, April 2019

Weather Classification using Machine Learning at the Telescope Array Cosmic Ray Observatory (Poster), Department of Physics and Astronomy Annual Symposium, University of Utah, Salt Lake City, UT, April 2019

Machine Learning Weather Classification with Fluorescence Detector Pedestal Data at the Telescope Array Cosmic Ray Observatory, APS Four Corners Meeting, University of Utah, Salt Lake City, UT, Oct. 2018

Machine Learning Weather Classification with Fluorescence Detector Pedestal Data at the Telescope Array Cosmic Ray Observatory, Graduate Research Seminar, University of Utah, Salt Lake City, UT, Oct. 2018

Dark Nights and Desert Air: Understanding the Highest Energy Particles in the Universe, Great Basin Astronomy Festival, Great Basin National Park, NV, Sept. 2018

TAx4 Fluorescence Detection (Poster), Physics and Astronomy Symposium, University of Utah, Salt Lake City, UT, April 2017

Hosting Your Own Website from Home with a Raspberry Pi, Graduate Research Seminar, University of Utah, Salt Lake City, UT, Oct. 2017

Once More Unto the Breach: Advice from a Physics graduate Student, Physics Seminar, Weber State University, Ogden, UT, Feb. 2017

Peer Mentoring: Training and Supporting Physics TAs at the University of Utah (Poster), AAPT Idaho-Utah Section Meeting, Pocatello, ID, April 2016

Preliminary Search for Exotic Events in the Pierre Auger Cosmic Ray Observatory Data, Physics Senior Research Oral Presentations, Michigan Technological University, Houghton, MI, April 2014

AWARDED GRANTS AND FUNDING

Received **25 Awards**: **$ 2.63 M total** funding, **13 unique** awards, **11 extensions**, with awards from over **11 sponsors**.

Aerospace Corporation Award Extension II, PI, **$100,000** — Sept 2026 – Aug 2027
Expeditionary SDA

Adaptive and Intelligent Space (AIS) - SF25D-T1201, Institutional PI (Prime: Citra Space Corporation), **$97,000** — Sept – Dec 2026
Resilient Edge Autonomy for Space Domain Awareness

Adaptive and Intelligent Space (AIS) - SF25D-T1201, Co-PI (Prime: RAIC Labs), **$22,500** — Sept – Dec 2026
Efficient AI/ML Models at the Edge

Lockheed Martin Corporation CTO University Research Program, PI, **$100,000** — Sept 2026 – Aug 2027
Closely Spaced Objects (CSO) Tracking

DoD USAF SBIR SF254-D1006 D2P2, Institutional Co-PI (Prime: Pulsar Space Systems), **$60,000** — Aug 2026
deTECt: GNSS based high-velocity object detection

Lockheed Martin Space Systems IRAD Extension VI, PI, **$46,000** — Jun 2026
Multi-Object Tracking

Lockheed Martin Space Systems IRAD Extension V, PI, **$99,000** — Jun 2026
Multi-Object Tracking

Lockheed Martin Space Systems IRAD Extension IV, PI, **$40,000** — Dec 2025
Multi-modal Dim Object Detection

Lockheed Martin Space Systems IRAD Extension III, PI, **$10,000** — Dec 2025
Multi-modal Dim Object Detection

SDA TAP Lab Cohort Award Extension II, Institutional PI (Prime: InTrack Radar Technology), **$80,000** — Aug 2025 – Dec 2025
Open-Source Launch Detection

AWARDED GRANTS AND FUNDING CONTINUED

Aerospace Corporation Award Extension, PI, **$90,000** — Sept 2025 – Aug 2026
SDA TAP Lab Student Projects

SDA TAP Lab Cohort Award, Institutional PI (Prime: GTC Analytics), **$25,000** — Mar 2025 – Mar 2026
Open Source Launch Detection

SDA TAP Lab Cohort Award, PI, **$50,000** — Jun – Aug 2025
SWIR Expeditionary SDA

Lockheed Martin Space Systems IRAD Extension II, PI, **$99,000** — Dec 2024
Multi-modal Dim Object Detection

Lockheed Martin Space Systems IRAD Extension I, PI, **$50,000** — Dec 2024
Multi-modal Dim Object Detection

SDA TAP Lab Cohort Award Extension I, Institutional PI (Prime: InTrack Radar Technology), **$80,000** — Oct 2024
Open-Source Launch Detection

DoD USAF SBIR SF243-D003, Institutional PI, **$347,660** — Nov 2025 – Jan 2026
SDA Target Detection and Characterization using WFOV Staring Data

AFRL ML-RCP POA-24-RV-001, Institutional PI, **$40,000** — Aug 2024 – Aug 2026
Grant Institution: Ohio State University, Partner Institution: Tennessee State University
Multimodality Sensor Data Synthesis and Fusion for Improved Space Domain Awareness (SDA)

AFX24D PTS01 STTR, Institutional PI (Prime: Exclosure), **$33,000**, — May – June 2024
Highest Revisit Frequency Space Situational Awareness Observatory

SDA TAP Lab Cohort Award, Institutional PI (Prime: InTrack Radar Technology), **$30,000** — Jun – Aug 2024
Open-Source Launch Detection

Aerospace Corporation Award, PI, **$80,000** — Sept 2024 – Aug 2025
SDA TAP Lab Student Projects

Lockheed Martin Space Systems IRAD, PI, **$50,000** — Jan – June 2024
Multi-modal Dim Object Detection

NSTXL OTA # FA8814-21-9-0001, Institutional PI (Prime: Lockheed Martin Space Systems), **$968,000** — Apr 2023 – Mar 2027
Overhead Persistent Infrared Data Exploitation Technology Transition, OPIR ODETT

Information Institute Visiting Faculty Research Program Extension, PI, **$10,000** — Oct 2023
Orbit-to-Tracklet Track-before-Detect Algorithm Development Extension

AFOSR Summer Faculty Fellowship Program, PI, **$24,240** — June – Sept 2023
Orbit-to-tracklet Hypothesis Constraints to Frame Shifting in a Track-Before-Detect Method for Wide Field of View Survey Telescopes

STUDENT MENTORING

Research Mentor, University of Colorado Boulder

Ike Olelewe, AES PhD Student — 2025 – Present
Transformer-based Time Sequence Characterization

Skye Crocker, Data Science Masters Student — 2025 – 2026
Multi-spectral Launch Detection with Weather Satellite Imagery

Copeland Laris, Data Science Masters Student, Graduated Spring 2026 — 2025 – 2026
Foreign Launch Detection with Weather Satellite Imagery

Tanner Gill, AES Masters Student, Graduating Fall 2026 — 2024 – 2026
RSO Characterization with Transformers

Hank Rains, AES Masters Student — 2024 – 2025
Similarity-based Clustering

Tim Behrer, AES Masters Student, Graduated Spring 2025 — 2024 – 2025
CNNs for Launch False Detection Control

STUDENT MENTORING CONTINUED

Research Mentor, University of Colorado Boulder

Pranav Chopdekar, Data Science Masters Student — 2024
Multi-spectral Object Detection

Joanna Wang, Computer Science Masters Student, Graduated Spring 2025 — 2024 - 2025
Multi-Modal Object Detection

Research Mentor, Lockheed Martin

Mridul Gupta, Purdue University Electrical and Computer Engineering, Defended PhD 2023 — 2021 – 2022
Thesis Advisor: Ed Delp, Video and Image Processing Laboratory (VIPER)
Sub-pixel localization and multi-spectral signal processing

United States Military Academy Cadet — 2022
Synthetic Image Generation Explainability

United States Military Academy Cadet — 2021 – 2022
Machine Learning Cloud Segmentation in Satellite Imagery

SERVICE

Center for National Security Initiatives, University of Colorado Boulder

Organizer, Seminar Series — 2023 – 2024

Academic Senate, University of Utah

Member, Graduate Assembly Ad Hoc Committee — 2019

College of Science, University of Utah

Member, College of Science Council — 2017 – 2018
Member, College of Science Curriculum Committee — 2017 – 2018
Member, College of Science College Student Council — 2017 – 2018

Department of Physics and Astronomy, University of Utah

Chair, Graduate Student Advisory Council — 2017 – 2018
Member, Graduate Student Advisory Council — 2015 – 2019

Science Outreach

Volunteer, Physics Open House, Weber State University — 2017, 2018, 2019
Volunteer, Science Day, University of Utah — 2017, 2019
Volunteer and Speaker, Great Basin Astronomy Festival, Great Basin NP — 2018

Society of Physics Students, Michigan Technological University Chapter

Treasurer — 2012 – 2013

MEDIA COVERAGE

Best Poster Award referenced in *The 2025 AMOS Conference Highlights Collaboration and Partnerships*, amostech.com, October 10, 2025

The Data Science Innovation Hub Powering the U.S. Space Force's Digital Defenses, Autonomy Global, Dawn Zoldi, August 27, 2025

Space Force Turns to TAP Labs To Bridge Tech Gaps, Payload, Tim Fernholz, October 8, 2024

Participation announced in *Game Changing U.S. Space Domain Awareness (SDA) at Space Systems Command*, USSF Space Systems Command (SSC) Public Affairs, El Segundo, CA, April 22, 2024

Participation annouced in *Startups, universities selected for accelerator focused on space domain awareness*, SpaceNews, Sandra Erwin, November 8, 2023

Award annouced in *Space Systems Command awards Data Exploitation Technology Transition I contract to Lockheed Martin*, USSF Space Systems Command (SSC) Media Release, El Segundo, CA, Oct. 2, 2023

Weather Classification using Machine Learning, University of Utah, Center for High Performance Computing, Research Highlights

Keeping the Student Voice: Student Involvement in the Tenure Process, Katelyn Collett, The Daily Utah Chronicle, May 22, 2019

Pictured in *Telescope Array TAx4 expansion work: Installation of surface particle detector begins in extreme cold* (translated), Makio Nakamura, ICRR News, **104**, Winter and Spring 2019

PhD candidate sheds light on work of Millard County cosmic-ray observatory that's largest in Northern Hemisphere, Rhett Wilkinson, Millard County Chronicle Progress, Sept. 19, 2018